



Fig. 2: Prompts that produced those images

advances even include speech and music creation. Zooming out into a timeline with some of the major generative AI milestones (Fig. 3), we see that this technology has been out for a while, but only gained monumental traction recently.

It all started with the paper titled ‘Improving Language Understanding by Generative Pre-Training’ released by OpenAI. The paper alluded to GPT (generative pretrained transformer), and laid the seed for ‘large language models.’ However, generative AI started coming into the limelight with the public release of GPT-3 and DALL-E. The former is a next-generation version of the already performant GPT-2 model and the latter is an image generative model like Stable Diffusion. Their public demos and API access set ablaze the imagination of technologists across the planet. However, these implementations remain closed-source and proprietary.

Programmers were enthused and drawn towards generative AI when GitHub opened up a private preview of GitHub Copilot. Copilot utilised OpenAI Codex—an extension of GPT-3 that is trained on thousands of GitHub repositories. This specialisation allowed it to generate code on-the-fly, infer func-

tionality of code and, more recently, automatically write test cases.

Around this time, EleutherAI created an open source model similar to GPT-3 as well with some impressive demos. It was named ‘GPT-J-6B’, the J alluding to the Google JAX framework used to develop the model.

However, nothing expanded the open source generative AI scene more than the open source release of Stability AI’s Stable Diffusion. Within hours of its release, technology forums were flooded with inspired individuals expanding the art of the possible. Several developers had built feature-complete, easy-to-use user interfaces and deployment strategies within days of its release. Some even released, modified, and optimised the model to suit specific use cases.

The release of Stable Diffusion was still not the biggest splash that generative AI had created in the world. That’s credited to the recent public demo release of ChatGPT, a purpose-built GPT-3 based model capable of holding context-aware conversations and providing an easy-to-use and familiar interface. It reached over one million users in the short span of a week, a figure that took earlier social networks months and years to achieve.

The keen-eyed among you

must have noticed that Stable Diffusion isn’t the first open source generative AI model that was launched publicly, yet it somehow broke out of the mould with millions of users. A large chunk of these users self-hosted this model, integrating it deeply into their products and open source projects. So,

what did they do differently?

The accessibility and wide adoption can be chalked down to two key factors.

Availability of pretrained weights

Generative AI models are often ‘deep neural networks’ with millions/billions of parameters, trained over data sets with hundreds and thousands of items, whether text or images. To look at some metrics, the cost of training GPT-3 is over 4.6 million dollars using a Tesla V100 cloud instance with training times of up to nine days. Similarly, Stable Diffusion was trained using 256 NVIDIA A100 GPUs on Amazon Web Services for a total of 150,000 GPU hours, at a cost of US\$600,000. These costs are prohibitive for individuals and even small businesses. Putting these trained models to use is more inexpensive by orders of magnitude.

Fortunately, it is possible to share the outcome of this training in the form of ‘weights,’ which can be loaded by anyone who wants to use the model. This is the approach taken by GPT-J 6B and Stable Diffusion, who had weights available for download from day one. This allows individuals that have access to sufficiently powerful hardware to load up the model and put it to use.